

## **MB 405: Database for Managers**

**Course Code: MB 405**

**Course Name: Database for Managers**

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: Integrated MBA

Mutual Exclusion: NA

Approval: 59th BoA (I-MBA curriculum); course proposal approved in the 65th BoA

### **Preamble**

This course introduces students to the fundamental concepts of business databases and their role in supporting organizational operations, analytics, and managerial decision-making. It explores the evolution of database technologies from traditional relational databases to modern graph databases, data warehouses, and data lakes. Students will learn how enterprise systems such as ERP, CRM, and SCM utilize databases to manage business information efficiently. The course covers relational database design, entity-relationship modeling, normalization, and SQL for business reporting and data retrieval. It also examines data warehousing, ETL/ELT processes, business intelligence, dashboards, and emerging data architectures including data lakehouses, cloud databases, data fabric, and data mesh. Additionally, students are introduced to graph databases and AI-enabled database technologies that support connected business ecosystems and real-time analytics. Emphasizing managerial applications rather than technical programming, the course equips students with the knowledge to understand, evaluate, and leverage modern data management technologies for strategic business decision-making in a data-driven economy.

Unlike traditional MBA database courses, this course integrates database fundamentals with strategic datagraph thinking, introduces relationship-centric business intelligence, connects databases with AI and Industry 4.0, and prepares students for modern connected enterprises. This course bridges classical database education with next-generation intelligent business ecosystems.

### **Objectives of the Course**

Students will be able to:

- Describe the evolution of database technologies, including relational, graph, and AI-enabled databases, and their applications in modern enterprises.
- Explain the role and importance of databases as strategic assets in supporting business operations and managerial decision-making.
- Apply fundamental concepts of relational database design, including entity-relationship modeling, normalization, keys, and relationships, to model business data.
- Use SQL to create, retrieve, manipulate, and analyze business data for reporting and decision support.
- Differentiate between operational databases, data warehouses, data lakes, and emerging data architectures such as data lakehouses, data fabric, and data mesh.
- Evaluate how enterprise systems, including ERP, CRM, and SCM, utilize databases to manage information and improve organizational performance.

- Explain the business applications of graph databases, connected data models, and real-time data processing in areas such as customer analytics, supply chain management, and fraud detection.
- Assess the impact of emerging technologies, including artificial intelligence, autonomous databases, and intelligent business systems, on future business data management and analytics.

### Course Modules

- **Introduction to Business Databases:** Databases as business assets- Business databases in organizations, Evolution of databases from hierarchical to graph, Databases in ERP, CRM, SCM systems, database and datagraph, Role of databases in managerial decision-making. (4 hours)
- **Relational Database Systems:** Database Management Systems (DBMS), Relational database concepts, Tables, rows, columns, Keys and relationships, Data integrity, and identifying business entities. Entity Relationship (ER) Models, Business process modeling, Relationship mapping, Normalization basics, Data redundancy, Designing business databases. (8 hours)
- **Data warehouse and data Lake:** Data Warehouse: Definition and Characteristics, Components of a Data Warehouse, Subject-Oriented Data, Data Warehouse Architecture, Fact Tables and Dimension Tables, Star Schema, Snowflake Schema, Data Marts. ETL Process: Data Cleaning, Integration and Quality, Metadata, OLAP vs OLTP, Dashboards, KPIs. Data Lakes: Definition and Characteristics, Schema-on-Read, ELT Process, Metadata Management, Data Governance, Data Swamps. Emerging Data Architecture: Data Lakehouse, Cloud Data Warehouses, Cloud Storage, Data Fabric and Data Mesh. (12 hours)
- **SQL for Managers:** SQL queries and Business report generation- Introduction to SQL, Data Definition Language (DDL), Data Manipulation Language (DML), Queries and filters, Sorting and grouping, Aggregate functions, Joins and relationships. (6 hours)
- **Graph Databases & Connected Business:** Introduction to graph databases, Nodes and relationships, Graph thinking for managers, Neo4j overview, Connected customer ecosystems. (6 hours)
- **AI, Real-Time Data & Future Databases:** AI-enabled databases, Data streaming, Digital twins, Intelligent business systems, Autonomous databases, Industrial AI, Fusion business models, Connected intelligence systems. (6 Hours)

### Laboratory/Practical/Tutorial Modules

None

### Text Books

1. Abraham Silberschatz, Henry F. Korth, and S. Sudarshan, *Database System Concepts*, McGraw-Hill.
2. Turban, E., Sharda, R., Delen, D., & King, D., *Business Intelligence, Analytics, and Data Science: A Managerial Perspective*. Pearson (latest edition).

### References

1. Ponniah, P., *Data Warehousing Fundamentals for IT Professionals*, Wiley, 2010.
2. Robinson, I., Webber, J., and Eifrem, E., *Graph Databases: New Opportunities for Connected Data*, O'Reilly Media, 2015.

3. Gosnell, D., & Broecheler, M., *The Practitioner's Guide to Graph Data*. O'Reilly Media, 2020.
4. Lee, V., Nguyen, P. K., & Thomas, A., *Graph-Powered Analytics and Machine Learning with TigerGraph: Driving Business Outcomes with Connected Data*. O'Reilly Media, 2023.
5. Kejriwal, M., Knoblock, C., & Szekely, P., *Knowledge Graphs: Fundamentals, Techniques, and Applications*. MIT Press, 2021.
6. Alex Gorelik, *The Enterprise Big Data Lake: Delivering the Promise of Big Data and Data Science*. O'Reilly Media, 2019.
7. Provost, F and Fawcett, T., *Data Science for Business*, Shroff Publishers and Distributors Pvt. Ltd, 2014.
8. Raghuram Krishnan & Johannes Gehrke, *Database Management Systems*. McGraw-Hill.
9. Bill Schmarzo, *Big Data: Understanding How Data Powers Big Business*. Wiley, 2013.
10. Kimball, R., & Ross, M, *The Data Warehouse Toolkit* (3rd ed.). Wiley, 2013.
11. DeBarros, A., *Practical SQL: A Beginner's Guide to Storytelling with Data*, No Starch Press, 2022.